

Hermitian (self-adjoint) operator $H^\dagger = H$

- Every Hermitian operator $H^\dagger = H$ has only real eigenvalues $\lambda^* = \lambda$, and $\langle v | H(v) \rangle = v^\dagger H v$ is also real.
- In quantum physics, they represent observables such as position, momentum, energy, and spin-components.
- Like every complex number $z = x + iy$, every operator $A = \underbrace{\frac{A + A^\dagger}{2}}_{A_x} + i \underbrace{\frac{A - A^\dagger}{2i}}_{A_y}$, where both $A_{x,y}$ are

Hermitian operators.

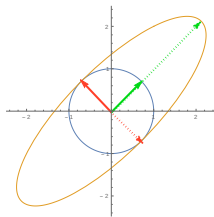
- **Real** linear combination of Hermitian operators $x H_x + y H_y$ is also a Hermitian operator.
- If H is a Hermitian operators, then $S = iH$ will be a Skew-Hermitian operator, $S^\dagger = -S$, whose eigenvalues will be purely imaginary numbers $\chi^* = -\chi$.
- Product of two Hermitian operators AB will not be a Hermitian operator unless they commute $AB = BA$.

Effect of H on the unit circle:

$$H e_1 = 3 e_1 \quad \text{and} \quad H e_2 = -e_2$$

$$e_1 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix} \quad \text{and} \quad e_2 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix}$$

$$H(\underbrace{\alpha_1 e_1 + \alpha_2 e_2}_v) = \alpha_1 \lambda_1 e_1 + \alpha_2 \lambda_2 e_2$$



video